

do nothing until you see a probability that is wildly mispriced. Option math works a lot better over short intervals. Once you extend the time horizon, all sorts of exogenous variables are introduced that can throw a wrench into the option-pricing model.

Another example of a distortion that is introduced when the time interval is extended relates to the fact that the option-pricing models assume that volatility increases with the square root of time. This assumption may provide reasonable approximations for shorter time intervals, say one year or under, but if you have a very low standard deviation, and you extend it for a very long time, it doesn't scale properly. For example, if a one-year standard deviation is 5 percent, assuming that the nine-year standard deviation will only be 15 percent is probably an underestimate.

I guess the reason is that the longer the time period, the greater the potential for a trend, and hence the greater the chance that a longer-term price move will exceed the standard deviation implied probability, which increases only by the square root of time.

Yes.

Wouldn't that in turn imply that any long-term option is likely to be priced too low?

We love long-term options.

All the trades we have discussed seem to share two common denominators: a mispricing that arises because standard market pricing assumptions are inappropriate for a given situation and an asymmetric return/risk profile—that is, open-ended return and curtailed risk. Are there any other common denominators in your trades?

For any trade idea we come up with, we always go in hoping that our work will lead us to table-pounding confidence about a directional view. When we find it, nothing is better. Unfortunately those situations happen only very rarely. Almost invariably, at some point along the way, we end up disproving one of the predicates to our hypothesis, or decide that the risk we are evaluating really is inscrutable, or that we overlooked some important factor, and the situation was never interesting to begin with.